

Development of Non-destructive Soil Bulk Density Measurement Technology for Efficient and Precise Estimation of Soil Carbon Stocks



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Materials and Methods details



Experimental Materials

- **Soil and Gravel Substrates**

Table 1. Physical properties of the experimental soil and gravel

Soil or Gravel	Land use	Bulk density(g/cm ³)	Soil texture ^a
NM	grass land	1.51 - 1.82	Sandy loam
SEO	wet land	1.23 - 1.60	Sandy loam
YM	grass land	1.31 - 1.61	Loam
SS	grass land	1.18 - 1.38	Sandy clay loam
Gravel ^b	-	1.41	Granite

All soil samples were sieved to pass a 2 mm mesh.

^aSoil texture was determined via the hydrometer method using 0.5% sodium hexametaphosphate as a dispersant.

^bGravel was sieved to isolate the 2–4 mm size fraction for experimental use.

- **Thermal Conductivity Analysis**



Fig. 1. Apparatus for thermal conductivity measurement

- Model: TEMPOS (METER Group)
- Sensor: SH-3 Dual-needle sensor

- **Experimental Pot Setup**

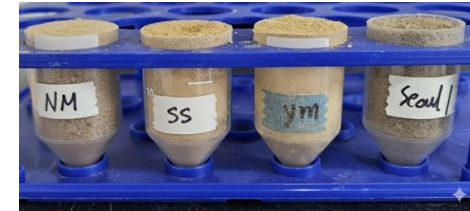


Fig. 2. Prepared sample specimen

- Samples were prepared using 20 mL conical tubes, modified from 50 mL stock.

Variable Manipulation and Experimental Methods



- **Gravel ratios**

- Reason for Selection: Current bulk density estimation models do not account for high gravel ratio in forest, grassland, and mountain soils.
- Manipulation Method: Fill 20 mL conical tubes by mixing gravel and soil (< 2 mm) at weight ratios of 0, 0.1, and 0.2.

- **Moisture Content**

- Reason for Selection: Moisture content significantly impacts in-situ soil bulk density.
- Studies show thermal conductivity methods have high explanatory power for moisture content (Tian, 2016).
- Manipulation Method: Fill 20 mL conical tubes by mixing water and soil+gravel at volume ratios of 0, 0.1, and 0.2.



Variable Manipulation and Experimental Methods

- **Sample Preparation and Measurement**

- Dry soil samples in an oven at 60°C for at least 4 hours
- Place calculated amounts of samples for each variable into 20 mL conical tubes
- Leave in a sealed space for at least 12 hours for temperature and moisture stabilization, and record moisture evaporated during this period
- Measure thermal conductivity using a thermal conductivity meter
(measured twice per sample; re-measure if the CV value between measurements is 5% or higher)

- **Final Variables**

- Total number of samples = 4 soil types × 3 gravel ratios × 3 moisture contents × 3 replicates = 108 samples
- Replicates per sample: 3 times

Method for estimating bulk density through thermal conductivity measurement



• Simplified de-Vries model

: A model to calculate the thermal conductivity within the soil space using the ratio of the three phases of soil composition, the thermal conductivity of each element, and weighting factors

$$\lambda_{\text{predict}} = \frac{\theta_{\text{water}} \lambda_{\text{water}} + k_{\text{air}} f_{\text{air}} \lambda_{\text{air}} + k_{\text{soil}} \lambda_{\text{soil}} \frac{\rho_b}{\rho_p}}{\theta_{\text{water}} + k_{\text{air}} f_{\text{air}} + k_{\text{soil}} \frac{\rho_b}{\rho_p}} \quad (\theta_{\text{water}} > 0) \quad (1)$$

$$\lambda_{\text{predict}} = 1.25 \times \frac{f_{\text{air}} \lambda_{\text{air}} + k_{\text{soil}} \lambda_{\text{soil}} \frac{\rho_b}{\rho_p}}{f_{\text{air}} + k_{\text{soil}} \frac{\rho_b}{\rho_p}} \quad (\theta_{\text{water}} = 0) \quad (2)$$

$$\lambda_{\text{soil}} = \lambda_{\text{sand}}^{f_{\text{sand}}} \lambda_{\text{silt}}^{f_{\text{silt}}} \lambda_{\text{clay}}^{f_{\text{clay}}} \quad (3)$$

$$k_n = \frac{2}{3} \left[1 + \left(\frac{\lambda_n}{\lambda_0} - 1 \right) g_{a(n)} \right]^{-1} + \frac{1}{3} \left[1 + \left(\frac{\lambda_n}{\lambda_0} - 1 \right) (1 - 2g_{a(n)}) \right]^{-1} \quad (4)$$

$$g_{a(\text{air})} = 0.333 \left(1 - \frac{f_{\text{air}}}{1 - \frac{\rho_b}{\rho_p}} \right) \quad (5)$$

$$g_{a(\text{soil})} = g_{a(\text{sand})} f_{\text{sand}} + g_{a(\text{silt})} f_{\text{silt}} + g_{a(\text{clay})} f_{\text{clay}} \quad (6)$$

θ_{water} = Volumetric ratio of water in soil (v/v%)

f_{air} = Volumetric ratio of air in soil (v/v%)

ρ_b / ρ_p = Volumetric ratio of particles in soil (v/v%)

ρ_b = Soil bulk density (g/cm³)

ρ_p = Soil particle density (2.65 g/cm³)

λ_{water} = Thermal conductivity of water in soil (0.596 W/m·K @ 20°C)

λ_{air} = Thermal conductivity of air in soil (0.025 W/m·K)

λ_{soil} = Thermal conductivity of soil particles

k_n = Weighting factor for component 'n' in soil

$\lambda_{\text{sand}}, \lambda_{\text{silt}}, \lambda_{\text{clay}}$ = Thermal conductivity of sand, silt, clay particles (7.70, 2.74, 1.93 W/m·K, respectively)

$f_{\text{sand}}, f_{\text{silt}}, f_{\text{clay}}$ = Mass fraction of sand, silt, clay particles

λ_0 = Thermal conductivity of the background medium (λ_{air} if $\theta_{\text{water}} = 0$, λ_{water} if $\theta_{\text{water}} > 0$)

$g_{a(n)}$ = de-Vries shape factor for soil component 'n'

$g_{a(\text{sand})}, g_{a(\text{silt})}, g_{a(\text{clay})}$ = de-Vries shape factor for sand, silt, clay (0.182, 0.0534, 0.00775, respectively)

Method for estimating bulk density through thermal conductivity measurement



- **GRG nonlinear Solver**

- A GRG nonlinear solver was used to find the bulk density that minimizes the Square Error between measured thermal conductivity ($\lambda_{\text{measured}}$) and predicted thermal conductivity (λ_{predict}) from the model.

- **Workflow of GRG Nonlinear Solver**

- Measure soil thermal conductivity ($\lambda_{\text{measured}}$)
- Set initial bulk density (1 g/cm³) in the model
- Calculate Square Error between $\lambda_{\text{measured}}$ and λ_{predict} (calculated from initial bulk density)
- Define Square Error as the objective function for the GRG Nonlinear Solver
- Find the bulk density value that minimizes the Square Error
- The final bulk density is calculated from the optimized volumetric water content and $\lambda_{\text{measured}}$

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Simplified de-Vries model improvement method to reflect the gravel ratio



- The particle-related variables were changed so that the Simplified de-Vries model could reflect the gravel ratio.
- For λ_{soil} gravel information was added to the geometric mean, which is composed of the ratio and thermal conductivity of particles less than 2 mm, eq(3.1).
- For λ_{gravel} , experimentally measured thermal conductivity data for domestic granite (Park et al., 2007) used in the experiment were used.
- For f_{sand} , f_{silt} and f_{clay} the ratios were reset considering the gravel ratio, which changed them to f'_{sand} , f'_{silt} , f'_{clay} and f'_{gravel} .
- For $g_{a(gravel)}$, since no literature information was available, the method used by (Tian et al., 2016) to derive $g_{a(silt)}$ was applied, and the value of 0.034 calculated using the GRG nonlinear solver method was used as $g_{a(gravel)}$, eq(6.1).

$$\lambda'_{soil} = \lambda_{sand}^{f'_{sand}} \lambda_{silt}^{f'_{silt}} \lambda_{clay}^{f'_{clay}} \lambda_{gravel}^{f'_{gravel}} \quad (3.1)$$

$$g'_{a(soil)} = g_{a(sand)} f'_{sand} + g_{a(silt)} f'_{silt} + g_{a(clay)} f'_{clay} + g_{a(gravel)} f'_{gravel} \quad (6.1)$$



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